Kraken2 × PathSeq Concordance — Full High-Resolution, Per-Sample

Same 5 cohorts and taxonomic-lineage rows as the cohort-aggregated figure, but every Kraken2/PathSeq pair is now split into one column-pair PER SAMPLE (thin dividers) within each cohort (thick dividers) — e.g. the CMV time course shows all 4 timepoints individually rather than summed into one column.

	CMV Fibroblast (4 samples)							
	12h		24h		48h		72h	
	PathSeq	Kraken2	PathSeq	Kraken2	PathSeq	Kraken2	PathSeq	Kraken2
Human CMV (HHV-5) *Cytomegalovirus humanbeta5 · 10359*	40.46% (643,233)	99.91% (1,588,558)	28.54% (1,133,300)	99.95% (3,969,423)	25.93% (1,515,321)	99.94% (5,841,101)	34.79% (690,672)	100.00% (1,985,167)
› Baboon/NHP CMV cross-map *Cytomegalovirus papiinebeta3 · 3050337 (K2) / 2169863*	n/a†	0% (0)	n/a†	0% (0)	n/a†	0% (0)	n/a†	0% (0)
› Cercopithecine CMV (bg) *Cytomegalovirus cercopithecinebeta5 · 3050258*	1.22e-04% (2)	0.0524% (833)	1.44e-04% (6)	0.0452% (1,795)	3.31e-05% (2)	0.0540% (3,154)	2.01e-05% (0)	0% (0)
› Chimp/Pan CMV (bg) *Cytomegalovirus paninebeta2 · 3050334*	0.0104% (165)	0.0181% (288)	0.0064% (253)	0.0059% (236)	0.0058% (338)	0.0036% (211)	0.0085% (169)	0.0030% (60)
HSV-1 (HHV-1) *Human alphaherpesvirus 1 · 10298*	0% (0)	0% (0)	5.04e-05% (2)	3.02e-04% (12)	1.71e-04% (10)	3.08e-04% (18)	4.53e-04% (9)	0.0011% (22)
› HSV-2 (specificity ctrl) *Human alphaherpesvirus 2 · 10310*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
VZV (HHV-3) *Human alphaherpesvirus 3 · 10335*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
› Bovine alphaherpesvirus 1 (ctrl) *Bovine alphaherpesvirus 1 · 10320*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
EBV (HHV-4) *Human gammaherpesvirus 4 · 10376*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
› Baboon/macaque EBV cross-map *Lymphocryptovirus/Macacine gamma-4 · 3050339 / 45455*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
HERV-K *Human endogenous retrovirus K · 45617 (same ID, both methods)*	0.0161% (256)	0.0170% (270)	0.0015% (58)	0.0026% (102)	5.82e-04% (34)	9.07e-04% (53)	4.03e-04% (8)	8.56e-04% (17)

	TG VZV/HSV1 Ganglia (5 donors)									
	TG3		TG4		TG5		TG6		TG7	
	PathSeq	Kraken2	PathSeq	Kraken2	PathSeq	Kraken2	PathSeq	Kraken2	PathSeq	Kraken2
Human CMV (HHV-5) *Cytomegalovirus humanbeta5 · 10359*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
› Baboon/NHP CMV cross-map *Cytomegalovirus papiinebeta3 · 3050337 (K2) / 2169863*	n/a†	0% (0)	n/a†	0% (0)	n/a†	0% (0)	n/a†	0% (0)	n/a†	0% (0)
› Cercopithecine CMV (bg) *Cytomegalovirus cercopithecinebeta5 · 3050258*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
› Chimp/Pan CMV (bg) *Cytomegalovirus paninebeta2 · 3050334*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
HSV-1 (HHV-1) *Human alphaherpesvirus 1 · 10298*	16.87% (1,464,210)	95.26% (8,267,423)	14.43% (1,816,718)	93.02% (11,707,977)	12.80% (2,251,920)	99.71% (17,540,230)	15.40% (7,677,324)	99.05% (49,372,617)	17.72% (1,910,916)	98.05% (10,575,993)
› HSV-2 (specificity ctrl) *Human alphaherpesvirus 2 · 10310*	5.76e-06% (0)	0% (0)	5.30e-06% (1)	0% (0)	5.02e-05% (9)	0% (0)	5.42e-05% (27)	0% (0)	1.55e-05% (2)	0% (0)
VZV (HHV-3) *Human alphaherpesvirus 3 · 10335*	0.598% (51,931)	4.713% (409,012)	0.748% (94,193)	6.968% (877,025)	0.0456% (8,026)	0.288% (50,682)	0.127% (63,180)	0.946% (471,534)	0.228% (24,633)	1.939% (209,184)
› Bovine alphaherpesvirus 1 (ctrl) *Bovine alphaherpesvirus 1 · 10320*	0% (0)	1.73e-04% (15)	0% (0)	0% (0)	1.71e-05% (3)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
EBV (HHV-4) *Human gammaherpesvirus 4 · 10376*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
› Baboon/macaque EBV cross-map *Lymphocryptovirus/Macacine gamma-4 · 3050339 / 45455*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
HERV-K *Human endogenous retrovirus K · 45617 (same ID, both methods)*	0.0019% (162)	0.0228% (1,977)	9.06e-04% (114)	0.0098% (1,232)	0.0034% (592)	0.0066% (1,163)	7.14e-04% (356)	0.0026% (1,286)	0.0017% (178)	0.0113% (1,221)

	EBV Lymphoblastoid (2 replicates)			
	Rep 1		Rep 2	
	PathSeq	Kraken2	PathSeq	Kraken2
Human CMV (HHV-5) *Cytomegalovirus humanbeta5 · 10359*	0% (0)	0% (0)	0% (0)	0% (0)
› Baboon/NHP CMV cross-map *Cytomegalovirus papiinebeta3 · 3050337 (K2) / 2169863*	0% (0)	0.0273% (71)	0% (0)	0.0473% (90)
› Cercopithecine CMV (bg) *Cytomegalovirus cercopithecinebeta5 · 3050258*	0% (0)	0% (0)	0% (0)	0% (0)
› Chimp/Pan CMV (bg) *Cytomegalovirus paninebeta2 · 3050334*	0% (0)	0% (0)	0% (0)	0% (0)
HSV-1 (HHV-1) *Human alphaherpesvirus 1 · 10298*	0% (0)	0% (0)	0% (0)	0% (0)
› HSV-2 (specificity ctrl) *Human alphaherpesvirus 2 · 10310*	0% (0)	0% (0)	0% (0)	0% (0)
VZV (HHV-3) *Human alphaherpesvirus 3 · 10335*	0% (0)	0% (0)	0% (0)	0% (0)
› Bovine alphaherpesvirus 1 (ctrl) *Bovine alphaherpesvirus 1 · 10320*	0% (0)	0% (0)	0% (0)	0% (0)
EBV (HHV-4) *Human gammaherpesvirus 4 · 10376*	60.01% (155,795)	96.31% (250,052)	62.24% (118,505)	97.13% (184,950)
› Baboon/macaque EBV cross-map *Lymphocryptovirus/Macacine gamma-4 · 3050339 / 45455*	0.0137% (36)	2.676% (6,948)	0.0136% (26)	1.503% (2,861)
HERV-K *Human endogenous retrovirus K · 45617 (same ID, both methods)*	0.376% (976)	0.982% (2,549)	0.608% (1,158)	1.320% (2,513)

	Iadorola TG HSV-1+ (3 donors)						Iadorola TG HSV-1− (2 donors)			
	TG3		TG12		TG4		TG13		TG2	
	PathSeq	Kraken2	PathSeq	Kraken2	PathSeq	Kraken2	PathSeq	Kraken2	PathSeq	Kraken2
Human CMV (HHV-5) *Cytomegalovirus humanbeta5 · 10359*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
› Baboon/NHP CMV cross-map *Cytomegalovirus papiinebeta3 · 3050337 (K2) / 2169863*	n/a†	0.821% (14)	n/a†	0.684% (16)	n/a†	0% (0)	n/a†	1.858% (18)	n/a†	1.509% (18)
› Cercopithecine CMV (bg) *Cytomegalovirus cercopithecinebeta5 · 3050258*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
› Chimp/Pan CMV (bg) *Cytomegalovirus paninebeta2 · 3050334*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
HSV-1 (HHV-1) *Human alphaherpesvirus 1 · 10298*	12.84% (219)	22.99% (392)	12.53% (293)	19.97% (467)	0.983% (9)	1.201% (11)	0% (0)	0% (0)	0% (0)	0% (0)
› HSV-2 (specificity ctrl) *Human alphaherpesvirus 2 · 10310*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
VZV (HHV-3) *Human alphaherpesvirus 3 · 10335*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
› Bovine alphaherpesvirus 1 (ctrl) *Bovine alphaherpesvirus 1 · 10320*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
EBV (HHV-4) *Human gammaherpesvirus 4 · 10376*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
› Baboon/macaque EBV cross-map *Lymphocryptovirus/Macacine gamma-4 · 3050339 / 45455*	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)	0% (0)
HERV-K *Human endogenous retrovirus K · 45617 (same ID, both methods)*	47.57% (811)	76.19% (1,299)	55.65% (1,301)	79.34% (1,855)	63.97% (586)	98.80% (905)	49.23% (477)	98.14% (951)	58.76% (701)	98.49% (1,175)

% of that SAMPLE's total Kraken2-positive reads (log scale): 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}

n/a

Normalization (changed from the cohort-aggregated figure): for each SAMPLE-column, % = (taxon's raw read count in that sample ÷ sum of Kraken2 raw reads across every taxon shown for that SAME sample) × 100, applied identically to PathSeq and Kraken2 — a per-sample denominator, not a per-cohort one, so composition differences between individual donors/timepoints are visible rather than pooled away.
Sample order (index-aligned with every taxon's value list, each independently verified — see this script's module docstring): cmv = 12h,24h,48h,72h; tg (Depledge) = TG3,TG4,TG5,TG6,TG7 (confirmed via direct ENA filereport lookup + internal cross-check against this cohort's own VZV max/min); ebv = Rep 1, Rep 2; Iadorola HSV-1+ = TG3,TG12,TG4; Iadorola HSV-1− = TG13,TG2.
All values are the same header-verified, fully cross-validated numbers as the cohort-aggregated full_highres figure — no new data gathering, purely a per-sample re-slice of already-confirmed totals.
† PathSeq has no taxon distinct from 10359 representing Kraken2's specific proxy-species artifact — the Human CMV row's own value already answers the equivalent question in every cohort it applies to.
Source: docs/pathseq_validation_results_2026-08-15.md, research/cmv_taxonomy_investigation.md.